

Contact

📍 Boston, MA

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Education

University of Central Florida

M.S., Intelligent Systems
2007

Thesis: *Learning Human Behavior from Observation for Gaming Applications*

B.S., Computer Engineering
2005

Publications

HiCAT 6: software control infrastructure and calibration
Proc. SPIE 10698, 2018

Moriarty, C., Brooks, K., Soummer, R., et al.

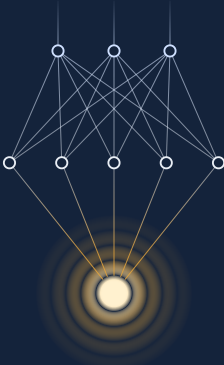
Learning human behavior from observation for gaming applications
Proc. FLAIRS-22, 439-444, 2009

Moriarty, C., Gonzalez, A. J.

Further instrumentation papers and abstracts

SPIE and AAS

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Christopher Moriarty

Senior Scientific Software Engineer

My work spans computer engineering, machine learning, and astrophysics, from space telescopes and planetary defense to radio interferometry. I want to bring that range back to science, and offer a professional software engineering perspective to the wild world of agentic coding, providing guardrails for safety, efficiency, and quality.

■ Experience 2024 to present

Moriarty Engineering LLC, Boston, MA · Owner/Contractor Oct 2024 – Present

Radio Observatory: Observing Modernization Design Contract Jul 2026 – Oct 2026

- Designing the automation of an interferometer’s observing, from proposal to archive, starting with rapid response to gamma-ray bursts: on target within minutes of an alert, with operator approval.
- Wrote the use cases with operators, PIs, and staff scientists, then the traceable requirements, a decision log, the System Architecture Document, and a phased roadmap for the build.
- Ran the two prototyping experiments behind the design with coding agents: OpenSpec proposals, agent-written changes on task branches, coding standards enforced by lint and unit tests in CI on every push, and my review of every merge.

Planetary Science Data Center: Infrastructure Contract Oct 2024 – Apr 2025

- Designed and deployed the primary virtualization system, XCP-ng and Xen Orchestra over a high-speed SAN, to move operations off aging infrastructure, with monitoring throughout.
- Migrated the core science database onto the new infrastructure, and built out a second physical site for backup and disaster recovery.

- AI Engineering and Self-Hosted Infrastructure
- Direct AI coding agents every day across my projects, on spec-first workflows (OpenSpec) with tests and my own review.
 - Built and run an all-Linux lab on two Dell PowerEdge servers: an XCP-ng virtualization host with Forgejo CI runners, git-driven Portainer deployments, and Prometheus, Grafana, and Loki monitoring, and a multi-GPU inference server that serves the open-weight models my agents run on, six vLLM streams across four GPUs at once.
 - Shipped two web apps this way: [Soundcheck](#), which gathers listings from dozens of Boston-area venues with playable tracks for each artist, and [JP Porchfest](#), an offline festival companion with on-device walking routes, built in a day, and used by hundreds of attendees.

■ Selected Projects

Observatory reference corpus

An observatory’s documentation, readable by agents

Built for the design contract. Gathers the observatory’s scattered websites, wikis, lecture decks, and memos into one Markdown corpus. Every page carries its source and date, an index routes questions, and a knowledge graph links the concepts. It settled most of the open design questions in a single session.

Markdown Python OpenSpec graphify

Science operations database

The new observing system’s database, tested on real nights

Built for the design contract. A new PostgreSQL schema for what to observe, what happened each night, and the calibrators. The package migrates the observatory’s full legacy record into it, replays real observing nights through it, and ships a review app for simulating observations interactively.

PostgreSQL Python Docker

Braid

Runs a coding agent from issue to pull request

A graph of spec-driven steps runs the agent. Each step must pass a scripted check on the repository, and a person approves the proposal and the merge. I built it to implement OpenSpec and Gitflow as graph workflows, and it is extensively tested.

Python Starlette React SQLite opencode

Trogdor

GPU inference server for my agents

I built the multi-GPU server by hand and engineered it to run quietly beside my desk. It hosts the open models my coding agents run on, through vLLM tuned for several agents at once. A one-line patch to vLLM made its older GPUs respond several times faster.

vLLM llama.cpp Docker Portainer Grafana

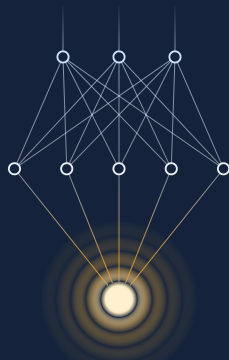
Skills

Languages AI and agents
Python Claude Code
C and C++ opencode
Java OpenSpec
SQL MCP
Bash vLLM
TypeScript llama.cpp

Web and data Infrastructure
SvelteKit Linux
React Docker
FastAPI XCP-ng
PostgreSQL iSCSI SAN
Redis Prometheus
RabbitMQ Grafana

Observatory systems
Real-time control systems
Data pipelines
Automated observing
Interferometry and VLBI
FPGA correlators
Coronagraph testbeds
Proposal and planning tools
Orbit determination
Archives and FITS

Engineering process
Software architecture
Requirements engineering
CI/CD and DevOps
Test-driven development
Code review
Project management
Technical leadership
Mentoring and training



■ Experience 2007 to 2024

Smithsonian Astrophysical Observatory, Cambridge, MA · Computer Engineer, GS-0854-14

Technical Manager, Minor Planet Center (MPC)

Sep 2021 – Aug 2024

- Led engineering for NASA's Planetary Defense clearinghouse of asteroid and comet observations, and reported regularly to NASA and the MPC Users Group.
- Hired and managed a team of engineers, and mentored an intern who built a near-Earth-object classifier in TensorFlow.
- Designed a parallel orbit-fitting pipeline: RabbitMQ and containerized OrbFit workers on Docker Swarm.
- Built public data visualization apps and Flask APIs on the MPC's observation schemas, on a replicated PostgreSQL cluster.
- Consolidated the code into a GitHub monorepo with feature branches, code review, and unit tests, and brought in Jira Service Desk to cut meeting overhead.
- Architected the on-prem infrastructure: a high-speed iSCSI SAN on Dell PowerVault arrays, XCP-ng virtualization, Prometheus and Grafana monitoring with alerting for automated fault detection, and a remote backup site.

Senior Software Engineer, Submillimeter Array

Sep 2018 – Sep 2021

- Led software for the interferometer on Maunakea, a station of the Event Horizon Telescope (EHT): improved networking and software for the FPGA-based SWARM correlator, and fixed and automated the VLBI software used in EHT campaigns.
- Created [smax-python](#), the Python client for SMA-X, the array's Redis-based real-time messaging layer. The SMA team still maintains it.
- Maintained and extended the array's C control software on LynxOS (a real-time operating system) and Linux, including the antenna sensor programs, corrPlotter, and dataCatcher, and bridged operator communications to Slack (2op2slack).
- Reworked the observatory's meetings and project management, and taught Python, testing, and version control in Cambridge and Hilo. That discipline carried SWARM's expansion to all six quadrants through a government shutdown, the Maunakea protests, and COVID.
- Founded and chaired the CfA Software Engineering Steering Committee, and co-organized a multi-day, observatory-wide operations review.

Space Telescope Science Institute, Baltimore, MD · Senior Systems Software Engineer

Lead Software Engineer, Makidon Optics Lab

Jan 2017 – Aug 2018

- Wrote the Python control library for HiCAT, the coronagraph testbed for NASA's next flagship telescope, with environmental sensing and safe shutdown that made remote operation possible. It was released as the open-source [catkit](#).
- Built the real-time data pipeline that wrote every exposure to FITS with complete metadata: background subtraction, bad-pixel correction, image registration, and archive-ready headers.
- Co-investigator on a funded NASA TDEM proposal, and first author of the testbed's software paper.

Astronomer's Proposal Tool (APT)

Aug 2012 – Jan 2017

- Led new features for the Java tool that scientists use to design and submit Hubble and JWST observing programs, gathering requirements from instrument scientists and building the JWST instrument templates.
- Recognized by NASA for bright-object checking of moving targets on Hubble and for JWST coordinated parallel observations.
- Built duplication checking against the MAST archive, kept the JUnit regression suite, and shepherded releases.

Harmonia, Blacksburg, VA · Software Engineer III

Aug 2010 – Jul 2012

- Led a Navy Phase II SBIR building AI-driven battle management software that drew on command-and-control systems to recommend resource plans, and won a follow-on contract to turn the code into an SDK.
- Applied neural networks, in C++ with the FANN library, to predict traffic patterns at a simulated intersection.

VIAVI (formerly JDSU), Raleigh, NC · Software Engineer I

Sep 2007 – Jul 2010

- Built device software in C and C++ for IPTV and network test equipment, from RS-232 control of a femtocell antenna and alarm filters and jitter measurement to a customer-facing XML API.
- Built the JBoss JSF monitoring and alarm system that showed live data from those devices, and took the product from formal requirements through deployment and customer support.